

Title: Mitochondrial DNA Variation in Migratory Bird Populations of the East Atlantic Flyway

Author: Thomas H. Bergstrom, Department of Zoology, University of Oxford

Abstract

Migratory birds along the East Atlantic Flyway experience strong selective pressures related to long-distance flight endurance. This study analyzes mitochondrial DNA sequences from twelve species of shorebirds to identify variants associated with metabolic efficiency during migration. We report a previously undescribed haplotype cluster in the cytochrome b gene that correlates with reduced stopover duration.

1. Introduction

The East Atlantic Flyway connects Arctic breeding grounds to sub-Saharan wintering sites, requiring some species to travel over 15,000 kilometers annually. Mitochondrial genes, which govern cellular respiration, are prime candidates for adaptations that support such extreme physiological demands.

2. Methods

Blood samples were collected from 340 individuals across twelve shorebird species at stopover sites in the Wadden Sea, the Banc d'Arguin, and the Wash. Mitochondrial genomes were sequenced and haplotype networks were constructed to examine geographic and phylogenetic structure.

3. Results

A distinct haplotype cluster, termed Cytb-7, was found at elevated frequency in individuals with above-average fat deposition rates prior to departure. Birds carrying this haplotype completed stopovers on average 1.8 days faster than those without it, suggesting a link between mitochondrial efficiency and refueling speed.

4. Discussion

These findings support the hypothesis that mitochondrial variation contributes meaningfully to migratory performance. Further work is needed to determine whether Cytb-7 confers a direct metabolic advantage or is linked to another functional locus through hitchhiking.